

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI





Dataset at a Glance

3,900

Purchases

18

Columns

\$59.76

Avg Spend

3.75

Avg Rating

Data Preparation in Python



→ **Clean**

Imputed missing Review Ratings;
standardized column names

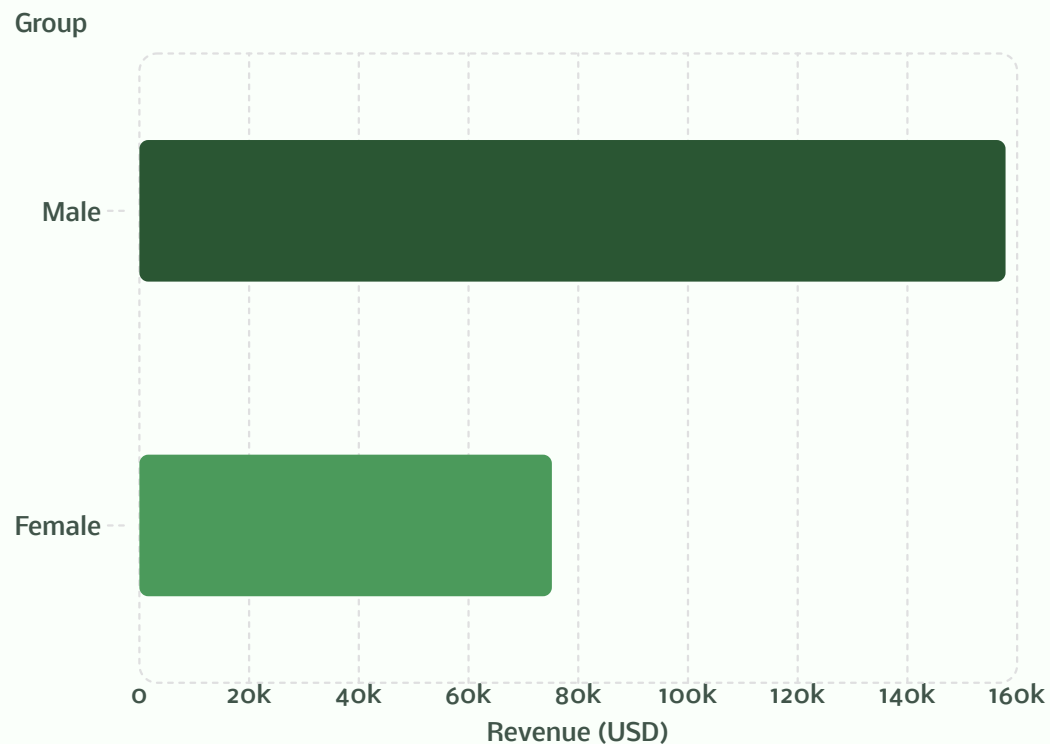
→ **Engineer**

Created `age_group` and
`purchase_frequency_days`

→ **Load**

Connected to PostgreSQL for
SQL analysis

Revenue by Gender & Age Group



Male customers generate 2× more revenue than female customers

Young-Adults lead age-group revenue at **\$62,143**, followed closely by Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763).

Customer Segmentation

Loyal

3,116 customers – 80% of base

Returning

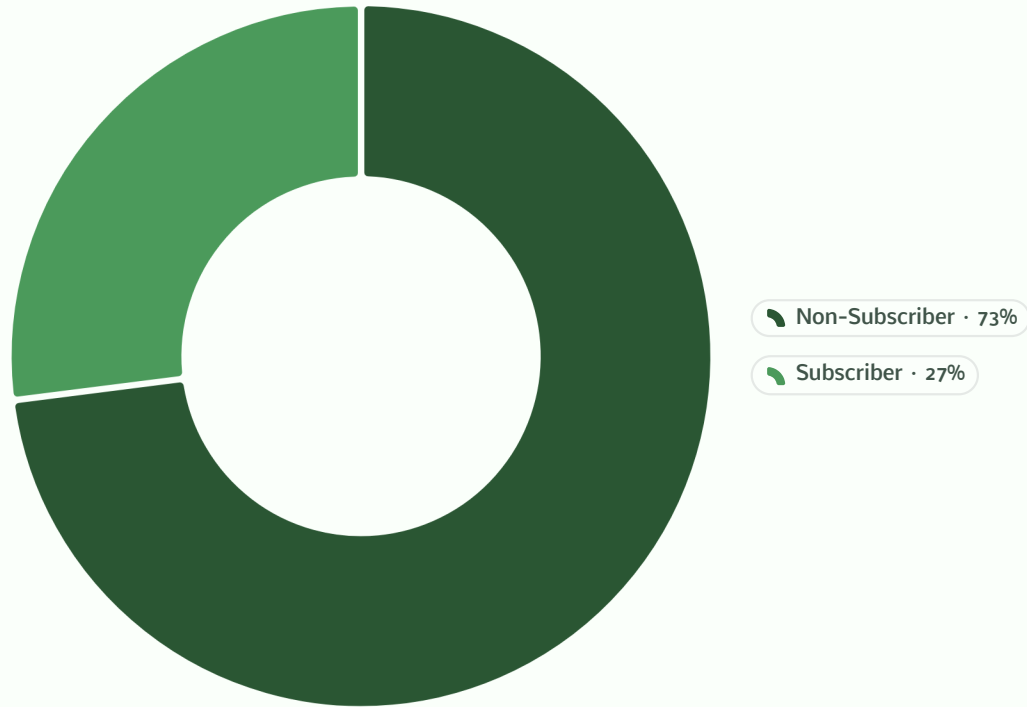
701 customers – growth opportunity

New

83 customers – focus for acquisition



Subscribers vs. Non-Subscribers



Only 27% are subscribers

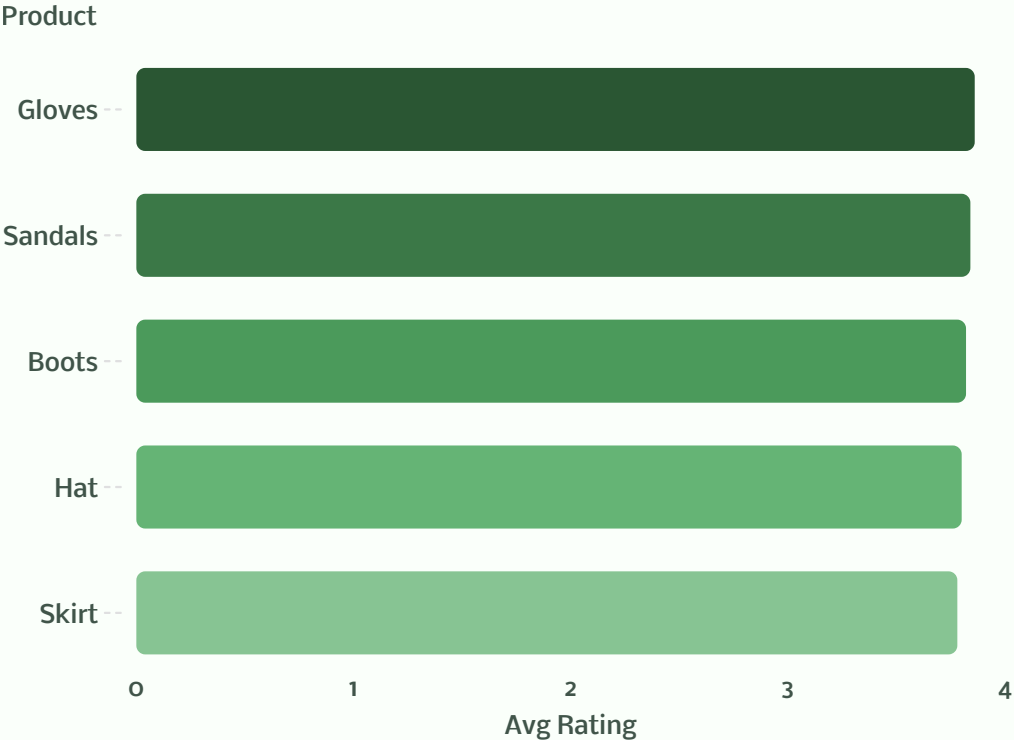
Non-subscribers: 2,847 customers · \$170,436 total revenue

Subscribers: 1,053 customers · \$62,645 total revenue

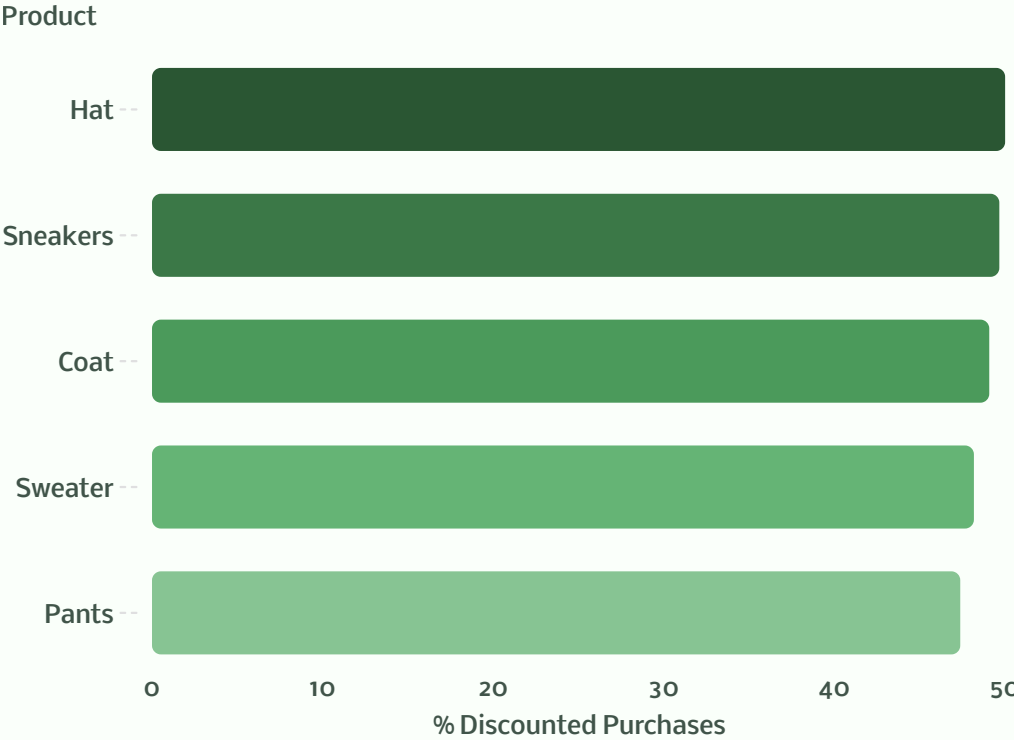
- ❏ Avg spend nearly identical: \$59.87 vs \$59.49 – subscription conversion is a low-hanging opportunity.

Top Products & Discount Behavior

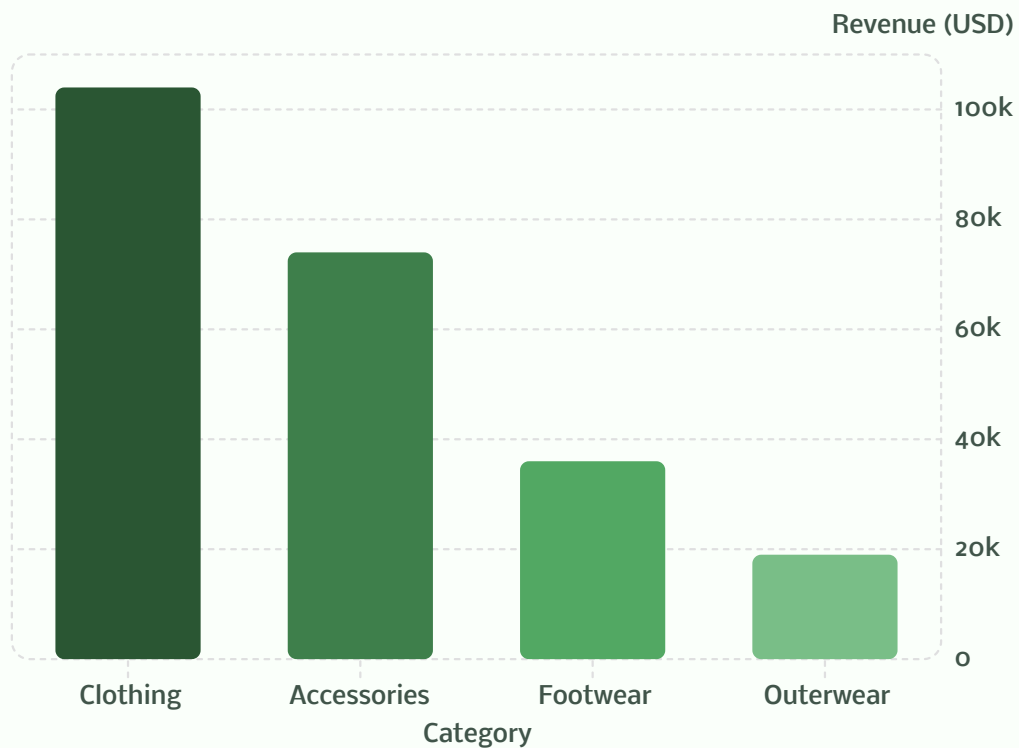
Top 5 by Rating



Most Discount-Dependent



Revenue by Category



Clothing dominates at \$104K

Top items: Blouse, Pants, Shirt (Clothing); Jewelry, Sunglasses, Belt (Accessories); Sandals (Footwear)

Express shipping avg: **\$60.48** vs Standard **\$58.46**

Power BI Dashboard

Filters

Subscription · Gender ·
Category · Shipping Type

KPIs

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Visuals

Revenue & sales by category and age group





Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users